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Vehicle and charging system

Abstract

A vehicle includes: a controller; a power receiving device that contactlessly receives power from a power transmitting facility; a battery chargeable using the power received by the power receiving device; and a power transmitting device that contactlessly outputs, to another vehicle, the power received by the power receiving device. The other vehicle contactlessly receives power from each of the power transmitting facility and the power transmitting device. When the other vehicle is unable to contactlessly receive the power from the power transmitting facility, the controller controls the power transmitting device to contactlessly output the power received by the power receiving device to the other vehicle in accordance with a request from the other vehicle.

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Background/Summary

(1) This nonprovisional application is based on Japanese Patent Application No. 2021-102301 filed on Jun. 21, 2021, with the Japan Patent Office, the entire contents of which are hereby incorporated by reference.

BACKGROUND

Field

(2) The present disclosure relates to a vehicle and a charging system, more particularly, a vehicle that contactlessly receives power and a charging system that contactlessly charges the vehicle with power.

Description of the Background Art

(3) Japanese Patent Laying-Open No 2017-143628 discloses a power transmitting system for

contactlessly transmitting power, to a vehicle stopped at an intersection to wait for traffic light, from a power transmitting facility provided at the intersection.

SUMMARY

(4) According to the power transmitting system disclosed in Japanese Patent Laying-Open No. 2017-143628, the battery of the vehicle can be charged using the power received contactlessly from the power transmitting facility provided at the intersection while the vehicle is stopped at the intersection to wait for traffic light. However, for example, when a large number of other vehicles are stopped to wait for traffic light, the foregoing vehicle may be unable to be stopped at a position at which power from the power transmitting facility can be received contactlessly, with the result that the battery may be unable to be charged using the power from the power transmitting facility.

(5) The present disclosure has been made to solve the above-described problem, and has an object to provide a technique of charging a battery of a vehicle using power from a power transmitting facility even when the vehicle cannot contactlessly receive the power from the power transmitting facility.

(6) A vehicle according to an aspect of the present disclosure is a vehicle that contactlessly receives power, and includes a controller, a power receiving device, a battery, and a power transmitting device. The power receiving device contactlessly receives power from a power transmitting facility. The battery is chargeable using the power received by the power receiving device. The power transmitting device contactlessly outputs, to another vehicle, the power received by the power receiving device. The other vehicle contactlessly receives power from each of the power transmitting facility and the power transmitting device. When the other vehicle is unable to contactlessly receive the power from the power transmitting facility, the controller controls the power transmitting device to contactlessly output the power received by the power receiving device to the other vehicle in accordance with a request from the other vehicle.

(7) A charging system according to an aspect of the present disclosure includes a first vehicle, a second vehicle, and a power transmitting facility. The power transmitting facility contactlessly transmits power to each of the first vehicle and the second vehicle. When the power from the power transmitting facility is unable to be contactlessly received, the first vehicle requests the second vehicle to output the power. When the power from the power transmitting facility is able to be contactlessly received, the second vehicle contactlessly outputs, to the first vehicle, the power received from the power transmitting facility in accordance with the request from the first vehicle.

(8) In the above configuration, when the other vehicle cannot contactlessly receive power from the power transmitting facility, the vehicle that can contactlessly receive power from the power transmitting facility outputs, to the other vehicle, the power received from the power transmitting facility in accordance with the request from the other vehicle. Thus, even when the vehicle cannot contactlessly receive the power from the power transmitting facility, the battery of the vehicle can be charged using the power from the power transmitting facility, thereby improving the user's convenience.

(9) The foregoing and other objects, features, aspects and advantages of the present disclosure will become more apparent from the following detailed description of the present disclosure when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a diagram showing exemplary charging points in a traveling route for a vehicle according to an embodiment.

(2) FIG. 2 is a diagram showing an exemplary configuration of the vehicle according to the embodiment.

- (3) FIG. 3 is a diagram showing an exemplary configuration of each of the vehicle and a power transmitting facility according to the embodiment.
- (4) FIG. 4 is a diagram showing an exemplary first charging system according to the embodiment.
- (5) FIG. 5 is a diagram showing an exemplary second charging system according to the embodiment.
- (6) FIG. 6 is a flowchart showing a process procedure of contactless charging between vehicles according to the embodiment.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (7) Hereinafter, embodiments of the present disclosure will be described in detail with reference to figures. It should be noted that in the figures, the same or corresponding portions are denoted by the same reference characters, and will not be described repeatedly.
- (8) <Exemplary Charging Points in Traveling Route for Vehicle>
- (9) FIG. 1 is a diagram showing exemplary charging points in a traveling route for a vehicle 1 according to an embodiment. FIG. 1 shows an example in which vehicle 1 travels from a departure point to a destination point via a rest point.
- (10) Vehicle 1 is an electrically powered vehicle having a battery 161 thereon for traveling, and is, for example, a battery electric vehicle (BEV) or a plug-in hybrid electric vehicle (PHEV). As shown in FIG. 1, an intersection 30 is provided in each of a traveling route R1 for vehicle 1 between the departure point and the rest point and a traveling route R2 for vehicle 1 between the rest point and the destination point. Intersection 30 is provided with a stop line 32 and a traffic light 33 for a traveling lane 31 leading to intersection 30.
- (11) In traveling lane 31, a power transmitting facility 9 is provided to precede stop line 32. Power transmitting facility 9 includes one or more power transmitting units 90. In this example, power transmitting facility 9 including four power transmitting units 90 is provided in traveling lane 31. It should be noted that power transmitting facility 9 may be provided in the ground of traveling lane 31 or may be provided on a side wall of traveling lane 31.
- (12) Power transmitting facility 9 employs each of power transmitting units 90 to contactlessly transmit power to vehicle 1 stopped at intersection 30 to wait for traffic light. Thus, battery 161 of vehicle 1 can be charged using the power contactlessly received from power transmitting facility 9 while vehicle 1 is stopped at intersection 30 to wait for traffic light.
- (13) A parking lot 40 is provided at the rest point such as a service area. Parking lot 40 includes: a parking space P1 provided with power transmitting facility 9; and a parking space P2 provided with no power transmitting facility 9. In this example, power transmitting facility 9 including one power transmitting unit 90 is provided in parking space P1. It should be noted that power transmitting facility 9 including a plurality of power transmitting units 90 may be provided in parking space P1. Further, power transmitting facility 9 may be provided in the ground of parking space P1 or may be provided on a side wall of parking space P1.
- (14) Power transmitting facility 9 employs power transmitting unit 90 to contactlessly transmit power to vehicle 1 parked in parking space P1. Thus, while vehicle 1 is parked in parking space P1, battery 161 can be charged using the power contactlessly received from power transmitting facility 9.
- (15) Thus, vehicle 1 can travel from the departure point to the destination point via the rest point with battery 161 being charged at the charging points such as intersection 30 and parking lot 40 by using the power contactlessly received from power transmitting facilities 9. It should be noted that in the description below, the charging that is contactlessly performed using power transmitting facility 9 is also referred to as “contactless charging”.
- (16) <Exemplary Configuration of Vehicle>
- (17) FIG. 2 is a diagram showing an exemplary configuration of the vehicle according to the embodiment. It should be noted that FIG. 2 shows a plurality of vehicles 1A, 1B each serving as vehicle 1 but both vehicles 1 have a similar configuration.

(18) As shown in FIG. 2, each of vehicles **1** includes an ECU (Electronic Control Unit) **11**, a navigation system **13**, and a communication module **14**. ECU **11**, navigation system **13**, and communication module **14** can communicate with one another via an in-vehicle network **15** such as a CAN (Controller Area Network).

(19) ECU **11** is an example of “controller” and includes: a processor such as a CPU (Central Processing Unit); and a memory **112** such as a ROM (Read Only Memory) and a RAM (Random Access Memory). Processor **111** is configured to perform a predetermined calculation process described in a program. Memory **112** stores a program to be executed by processor **111**. Further, memory **112** temporarily stores data generated by execution of a program by processor **111** and data input via communication module **14**. ECU **11** controls each device in vehicle **1** to bring vehicle **1** into a desired state based on a detected value of each of various types of sensors (not shown) and a program stored in memory **112**. Further, ECU **11** generates various types of data to be exchanged with each of other vehicle **1** and a server **2** via a network **20**.

(20) Navigation system **13** guides the traveling route for vehicle **1**. Navigation system **13** includes a GPS (Global Positioning System) receiver **131**, a display **132**, an operation unit **133**, a processor **134**, and a memory **135**.

(21) GPS receiver **131** specifies the location of vehicle **1** based on electric wave from an artificial satellite (not shown). Processor **134** performs various types of navigation processes (such as searching for a traveling route) based on the location information (GPS information) of vehicle **1** specified by GPS receiver **131**.

(22) Display **132** displays images (for example, icons) indicating various information. For example, display **132** displays an image indicating that contactless charging is being performed when vehicle **1** is being contactlessly charged, and displays an image indicating that contactless charging is not being performed when vehicle **1** is not being contactlessly charged.

(23) Operation unit **133** is constituted of a touch panel or the like, and receives various operations of a user (for example, a driver). For example, operation unit **133** receives an operation by the user to select a traveling route for vehicle **1**.

(24) Communication module **14** is an in-vehicle DCM (Data Communication Module), and is configured to allow for bidirectional communication between ECU **11** and each of other vehicle **1** and server **2** via network **20**.

(25) Server **2** manages the plurality of vehicles **1**. Further, server **2** holds the latest road information and holds the latest information regarding power transmitting facilities **9**. For example, server **2** collects the location information (GPS information) of each of the plurality of vehicles **1** and information on an amount of power (hereinafter, also referred to as “SOC” (State Of Charge)) stored in battery **161**. These pieces of information are periodically transmitted from each vehicle **1** to server **2**. Server **2** stores the collected information into a vehicle information database. Server **2** stores information on specifications (for example, charging method and charging capability), installation locations, and the like of power transmitting facilities **9**. Further, server **2** may acquire, from a controller (not shown) of traffic light **33**, information by which the time of waiting for traffic light **33** provided at intersection **30** can be specified.

(26) <Configuration Regarding Contactless Charging>

(27) FIG. 3 is a diagram showing exemplary configurations of vehicle **1** and power transmitting facility **9** according to the embodiment. In addition to the configuration shown in FIG. 2, vehicle **1** further includes battery **161**, an inverter **162**, a converter **163**, a motor generator **164**, power receiving devices **171** to **173**, and power transmitting devices **181**, **182**.

(28) Battery **161** is a battery pack including a plurality of cells. Each of the cells is a secondary battery such as a lithium ion battery or a nickel-metal hydride battery. Battery **161** supplies, to motor generator **164** via inverter **162**, power for generating driving force of vehicle **1**. Further, battery **161** stores power generated by motor generator **164**. Battery **161** is provided with a voltage sensor and a current sensor (both not shown) to allow ECU **11** to calculate the SOC of battery **161**.

(29) Inverter **162** converts DC power stored in battery **161** into AC power in accordance with control of ECU **11**, and supplies the AC power to motor generator **164**. Further, inverter **162** converts AC power (regenerative power) from motor generator **164** into DC power in accordance with control of ECU **11**, and charges battery **161** with the DC power.

(30) Converter **163** converts AC power received by each of power receiving devices **171** to **173** into DC power in accordance with control of ECU **11**, and charges battery **161** with the DC power. Converter **163** converts the DC power stored in battery **161** into AC power in accordance with control of ECU **11**, and outputs the AC power from each of power transmitting devices **181**, **182** to other vehicle **1**.

(31) Motor generator **164** receives power from inverter **162** and applies rotational force to the driving wheels, thereby causing vehicle **1** to travel.

(32) Power receiving device **171** is disposed, for example, at a lower portion of vehicle **1**. Power receiving device **171** includes a power receiving coil **1710**, and employs power receiving coil **1710** to contactlessly receive power output from power transmitting facility **9** provided in each of traveling lane **31** of intersection **30** and parking space **P1** of parking lot **40**.

(33) Power receiving device **172** is disposed, for example, at a side surface portion of vehicle **1**. Power receiving device **172** includes a power receiving coil **1720**, and employs power receiving coil **1720** to contactlessly receive power output from other vehicle **1** stopped beside the side surface of vehicle **1**.

(34) Power receiving device **173** is disposed, for example, at a front portion of vehicle **1**. Power receiving device **173** includes a power receiving coil **1730**, and employs power receiving coil **1730** to contactlessly receive power output from other vehicle **1** stopped in front of vehicle **1**.

(35) It should be noted that power receiving devices **171** to **173** may be integrated into one power receiving device. That is, vehicle **1** may employ one power receiving device to contactlessly receive the power output from power transmitting facility **9** or to contactlessly receive the power output from other vehicle **1** stopped beside the side surface of vehicle **1** or stopped in front of vehicle **1**.

(36) Power transmitting device **181** is disposed, for example, at a rear portion of vehicle **1**. Power transmitting device **181** includes a power transmitting coil **1810**, and employs power transmitting coil **1810** to contactlessly output power to other vehicle **1** stopped behind vehicle **1**. It should be noted that power transmitting device **181** is an example of “power transmitting device”; however, a device in which converter **163** and power transmitting device **181** are integrally formed may be the example of “power transmitting device”.

(37) Power transmitting device **182** is disposed, for example, at the side surface portion of vehicle **1**. Power transmitting device **182** includes a power transmitting coil **1820**, and employs power transmitting coil **1820** to contactlessly output power to other vehicle **1** stopped beside the side surface of vehicle **1**. It should be noted that power transmitting device **182** is an example of “power transmitting device”; however, a device in which converter **163** and power transmitting device **182** are integrally formed may be the example of “power transmitting device”.

(38) It should be noted that power transmitting devices **181**, **182** may be integrated into one power transmitting device. That is, vehicle **1** may employ one power transmitting device to contactlessly output power to other vehicle **1** stopped behind vehicle **1** or beside the side surface of vehicle **1**.

(39) Power transmitting facility **9** includes one or more power transmitting units **90** and a controller **95**. FIG. 3 shows an example in which power transmitting facility **9** includes four power transmitting units **90**. For example, as shown in FIG. 1, four power transmitting units **90** are disposed in a row in traveling lane **31** of intersection **30**. It should be noted that as shown in FIG. 1, one power transmitting unit **90** may be disposed in parking space **P1** of parking lot **40**.

(40) Each of power transmitting units **90** includes a power transmitting coil **910**. Power transmitting coil **910** is electrically connected to an AC power supply (not shown). Further, power transmitting unit **90** is provided with a sensor (an optical sensor, a weight sensor, or the like) (not

shown) for detecting vehicle **1** (or power receiving device **171**) located within a range in which contactless charging can be performed.

(41) Controller **95** specifies the location of vehicle **1** (or power receiving device **171**) based on a detection signal from each of the sensors described above. Then, controller **95** supplies AC power from the AC power supply, to power transmitting coil **910** of power transmitting unit **90** located within the range in which contactless charging to vehicle **1** can be performed.

(42) For example, when controller **95** detects that vehicle **1** is present above power transmitting unit **90** in traveling lane **31** or parking space **P1**, controller **95** supplies AC power to power transmitting coil **910**. Thus, alternating current flows through power transmitting coil **910**, with the result that an electromagnetic field is formed around power transmitting coil **910**. Power receiving coil **1710** in power receiving device **171** of vehicle **1** contactlessly receives the power through the electromagnetic field formed around power transmitting coil **910**. Thereafter, when vehicle **1** is no longer detected to be present above power transmitting unit **90** in traveling lane **31** or parking space **P1**, controller **95** stops supplying the AC power to power transmitting coil **910**. By performing such a series of control in one or each of the plurality of power transmitting units **90**, power transmitting facility **9** can contactlessly transmit power to vehicle **1**.

(43) <Contactless Charging Between Vehicles at Intersection>

(44) As described above, vehicle **1** is configured to contactlessly receive power from power transmitting facility **9** provided at intersection **30**. That is, when vehicle **1** can be stopped at a position at which power from power transmitting facility **9** can be contactlessly received while waiting for traffic light at intersection **30**, contactless charging from power transmitting facility **9** can be performed. However, for example, when a large number of other vehicles **1** are stopped to wait for traffic light, vehicle **1** may be unable to be stopped at the position at which the power from power transmitting facility **9** can be received contactlessly, with the result that battery **161** may be unable to be charged using the power from power transmitting facility **9**.

(45) Thus, vehicle **1** according to the embodiment is configured such that battery **161** can be charged using the power from power transmitting facility **9** through contactless charging from other vehicle **1** even when the power from power transmitting facility **9** cannot be received contactlessly. Hereinafter, a first charging system **101** according to the embodiment will be described.

(46) FIG. **4** is a diagram showing an exemplary first charging system **101** according to the embodiment. First charging system **101** includes a vehicle **1A**, a vehicle **1B**, and a power transmitting facility **9** provided at intersection **30** and configured to contactlessly transmit power to each of vehicle **1A** and vehicle **1B**. Vehicle **1A** is an example of “first vehicle”. Vehicle **1B** is an example of “second vehicle”.

(47) In traveling lane **31** of intersection **30**, the plurality of vehicles **1**, i.e., vehicles **1A** to **1C** are stopped to wait for traffic light. Each of vehicle **1C** located at the first place of the queue in waiting for traffic light and vehicle **1B** located immediately after vehicle **1C** is stopped at a position at which power from power transmitting facility **9** provided in traveling lane **31** can be received contactlessly. Thus, battery **161** of each of vehicle **1B** and vehicle **1C** can be charged using the power from power transmitting facility **9** while waiting for traffic light.

(48) On the other hand, vehicle **1A** located immediately after vehicle **1B** is not stopped at a position at which power from power transmitting facility **9** provided in traveling lane **31** can be received contactlessly. Therefore, battery **161** of vehicle **1A** cannot be charged using the power from power transmitting facility **9** while waiting for traffic light.

(49) Therefore, when the power from power transmitting facility **9** cannot be received contactlessly, vehicle **1A** requests vehicle **1B** located in front of vehicle **1A** to output the power. When the power from power transmitting facility **9** can be received contactlessly, vehicle **1B** contactlessly outputs, to vehicle **1A**, the power received from power transmitting facility **9** in accordance with the request from vehicle **1A**.

(50) Specifically, when ECU **11** of vehicle **1A** determines that power receiving device **171** cannot

receive the power from power transmitting facility **9**, ECU **11** outputs a request signal for requesting vehicle **1B** in front of vehicle **1A** to transmit power in accordance with an instruction input by the user of vehicle **1A** under conditions that a power transmission request condition is satisfied.

(51) The power transmission request condition includes, for example, a condition that a distance between vehicle **1A** and vehicle **1B** is less than or equal to a distance in which power can be contactlessly transmitted from vehicle **1B** to vehicle **1A**. That is, the power transmission request condition includes a condition that the distance between vehicle **1A** and vehicle **1B** falls within a range in which contactless charging can be performed.

(52) Upon receipt of the request signal from vehicle **1A**, vehicle **1B** outputs a permission signal for permitting to transmit power to vehicle **1A** having made the request in accordance with an instruction input by the user of vehicle **1B** under conditions that a power transmission permission condition is satisfied.

(53) The power transmission permission condition includes, for example, a condition that vehicle **1B** is contactlessly receiving power from power transmitting facility **9**, and/or a condition that the SOC of vehicle **1B** on the power transmitting side is larger than the SOC of vehicle **1A** on the power receiving side. It should be noted that the power transmission permission condition may include a condition that the SOC of vehicle **1B** on the power transmitting side has reached an upper limit value (for example, fully charged). Further, the power transmission permission condition may include a condition that the time of waiting for traffic light at intersection **30** is more than or equal to a threshold value. For example, vehicle **1B** may acquire, from server **2**, a signal by which the time of waiting for traffic light **33** provided at intersection **30** can be specified, and when the time of waiting for traffic light **33** is more than or equal to a time (threshold value) during which contactless charging from other vehicle **1A** can be performed, vehicle **1B** may output a permission signal for permitting to transmit power to vehicle **1A** having made the request in accordance with an instruction input by the user of vehicle **1B**.

(54) When vehicle **1A** receives the permission signal from vehicle **1B**, vehicle **1A** performs a power reception starting process for starting to contactlessly receive power from vehicle **1B**. Thus, vehicle **1B** contactlessly outputs, to vehicle **1A**, the power received from power transmitting facility **9** provided at intersection **30**. More specifically, ECU **11** of vehicle **1B** indirectly controls power transmitting device **181** via converter **163** so as to contactlessly output, to vehicle **1A**, the power received by power receiving device **171** from power transmitting facility **9**.

(55) On this occasion, battery **161** of vehicle **1B** may be charged using the power received by power receiving device **171** from power transmitting facility **9**, and the power stored in battery **161** may be contactlessly output to vehicle **1A**. Alternatively, vehicle **1B** may contactlessly output the power received by power receiving device **171** to vehicle **1A** directly without charging battery **161** using the power received by power receiving device **171** from power transmitting facility **9**. Further, vehicle **1B** may output, to vehicle **1A**, power corresponding to an amount of power contactlessly received from power transmitting facility **9** provided at intersection **30**.

(56) Thus, even when vehicle **1A** cannot contactlessly receive the power from power transmitting facility **9** provided in traveling lane **31** of intersection **30**, the power output from power transmitting device **181** provided at the rear portion of vehicle **1B** can be received contactlessly by power receiving device **173**, and battery **161** can be charged using the received power. Therefore, the user's convenience is improved.

(57) <Contactless Charging Between Vehicles in Parking Lot>

(58) As described above, vehicle **1** is configured to contactlessly receive the power of power transmitting facility **9** provided in parking lot **40**. That is, when vehicle **1** can be parked in parking lot **40** at parking space **P1** provided with power transmitting facility **9**, contactless charging from power transmitting facility **9** can be performed. For example, however, when other vehicle **1** is already parked in parking space **P1** provided with power transmitting facility **9**, vehicle **1** cannot

contactlessly receive power from power transmitting facility **9**.

(59) Thus, vehicle **1** according to the embodiment is configured such that battery **161** can be charged using the power from power transmitting facility **9** through contactless charging from other vehicle **1** even when the power from power transmitting facility **9** cannot be received contactlessly. Hereinafter, a second charging system **102** according to the embodiment will be described.

(60) FIG. **5** is a diagram showing an exemplary second charging system **102** according to the embodiment. Second charging system **102** includes a vehicle **1A**, a vehicle **1B**, and a power transmitting facility **9** provided in parking lot **40** and configured to contactlessly transmit power to each of vehicle **1A** and vehicle **1B**. Vehicle **1A** is an example of “first vehicle” Vehicle **1B** is an example of “second vehicle”.

(61) Vehicle **1B** is parked in parking space **P1** of parking lot **40**. Therefore, while vehicle **1B** is parked therein, battery **161** of vehicle **1B** can be charged using the power from power transmitting facility **9**.

(62) On the other hand, vehicle **1A** is parked in parking space **P2** provided with no power transmitting facility **9**. Therefore, while vehicle **1A** is parked therein, battery **161** of vehicle **1A** cannot be charged using the power from power transmitting facility **9**.

(63) Therefore, when the power from power transmitting facility **9** cannot be received contactlessly, vehicle **1A** requests adjacent vehicle **1B** to output power. When the power from power transmitting facility **9** can be received contactlessly, vehicle **1B** contactlessly outputs, to vehicle **1A**, the power received from power transmitting facility **9** in accordance with the request from vehicle **1A**.

(64) Specifically, when ECU **11** of vehicle **1A** determines that power receiving device **171** cannot receive power from power transmitting facility **9**, ECU **11** outputs a request signal for requesting adjacent vehicle **1B** to transmit power in accordance with an instruction input by the user of vehicle **1A** under conditions that a power transmission request condition is satisfied.

(65) The power transmission request condition includes, for example, a condition that a distance between vehicle **1A** and vehicle **1B** is less than or equal to a distance in which power can be contactlessly transmitted from vehicle **1B** to vehicle **1A**. That is, the power transmission request condition includes a condition that the distance between vehicle **1A** and vehicle **1B** falls within a range in which contactless charging can be performed.

(66) Upon receipt of the request signal from vehicle **1A**, vehicle **1B** outputs a permission signal for permitting to transmit power to vehicle **1A** having made the request in accordance with an instruction input by the user of vehicle **1B** under conditions that a power transmission permission condition is satisfied.

(67) The power transmission permission condition includes, for example, a condition that vehicle **1B** is contactlessly receiving power from power transmitting facility **9**, and/or a condition that the SOC of vehicle **1B** on the power transmitting side is larger than the SOC of vehicle **1A** on the power receiving side. It should be noted that the power transmission permission condition may include a condition that the SOC of vehicle **1B** on the power transmitting side have reached an upper limit value (for example, fully charged).

(68) When vehicle **1A** receives the permission signal from vehicle **1B**, vehicle **1A** performs a power reception starting process for starting to contactlessly receive power from vehicle **1B**. Thus, vehicle **1B** contactlessly outputs, to vehicle **1A**, the power received from power transmitting facility **9** provided in parking lot **40**. More specifically, ECU **11** of vehicle **1B** indirectly controls power transmitting device **182** via converter **163** so as to contactlessly output, to vehicle **1A**, the power received by power receiving device **171** from power transmitting facility **9**.

(69) On this occasion, battery **161** of vehicle **1B** may be charged using the power received by power receiving device **171** from power transmitting facility **9**, and the power stored in battery **161** may be contactlessly output to vehicle **1A**. Alternatively, vehicle **1B** may contactlessly output the power received by power receiving device **171** to vehicle **1A** directly without charging battery **161** using the power received by power receiving device **171** from power transmitting facility **9**.

Further, vehicle **1B** may contactlessly output, to vehicle **1A**, power corresponding to an amount of power contactlessly received from power transmitting facility **9** provided in parking lot **40**.

(70) Thus, even when vehicle **1A** cannot contactlessly receive the power from power transmitting facility **9** provided in parking lot **40**, the power output from power transmitting device **182** provided at the side surface portion of vehicle **1B** can be received contactlessly by power receiving device **172**, and battery **161** can be charged using the received power. Therefore, the user's convenience is improved.

(71) <Process Flow>

(72) FIG. **6** is a flowchart showing a process procedure of contactless charging between vehicles according to the embodiment. That is, FIG. **6** shows a procedure of each of the process between vehicle **1A** and vehicle **1B** in first charging system **101** of FIG. **4** and the process between vehicle **1A** and vehicle **1B** in second charging system **102** of FIG. **5**.

(73) The flowchart of FIG. **6** is invoked from a main routine (not shown) and executed when a predetermined condition is satisfied, for example. In the figure, the process performed by ECU **11** of the power receiving side vehicle is shown on the left side, and the process performed by ECU **11** of the power transmitting side vehicle is shown on the right side. The power receiving side vehicle corresponds to each of vehicle **1A** in first charging system **101** of FIG. **4** and vehicle **1A** in second charging system **102** of FIG. **5**. The power transmitting side vehicle corresponds to each of vehicle **1B** in first charging system **101** of FIG. **4** and vehicle **1B** in second charging system **102** of FIG. **5**. Each step is implemented by software processing by ECU **11** of each of the power receiving side vehicle and the power transmitting side vehicle, but may be implemented by hardware such as an LSI (Large Scale Integration) disposed in ECU **11**. Hereinafter, the term “step” is abbreviated as S.

(74) As shown in FIG. **6**, the power receiving side vehicle determines whether or not power can be contactlessly received from power transmitting facility **9** provided at intersection **30** or parking lot **40** (S**11**). For example, the power receiving side vehicle determines whether or not the power receiving side vehicle is located at a position at which power from power transmitting facility **9** can be received contactlessly. When the power receiving side vehicle can contactlessly receive the power from power transmitting facility **9** (YES in S**11**), the power receiving side vehicle returns the process to the main routine without performing contactless charging between the vehicles.

(75) On the other hand, when the power receiving side vehicle cannot contactlessly receive the power from power transmitting facility **9** (NO in S**11**), the power receiving side vehicle determines whether or not the power transmission request condition is satisfied (S**12**). When the power transmission request condition is not satisfied (NO in S**12**), the power receiving side vehicle returns the process to the main routine without performing contactless charging between the vehicles.

(76) On the other hand, when the power transmission request condition is satisfied (YES in S**12**), the power receiving side vehicle performs a process of displaying a power transmission request icon for receiving a power transmission request from the user (S**13**). For example, ECU **11** of the power receiving side vehicle displays the power transmission request icon on display **132** of navigation system **13**. Alternatively, ECU **11** of the power receiving side vehicle transmits an image signal for displaying the power transmission request icon to the user's mobile terminal (not shown) such as a smartphone via communication module **14**. Thus, the power transmission request icon is displayed on the display of the user's mobile terminal.

(77) The power receiving side vehicle determines whether or not the power transmission request by the user has been received (S**14**). For example, ECU **11** of the power receiving side vehicle determines whether or not an instruction to make the power transmission request by the user using operation unit **133** to operate the power transmission request icon displayed on display **132** of navigation system **13** has been received from navigation system **13**. Alternatively, ECU **11** of the power receiving side vehicle determines whether or not an instruction to make the power transmission request by the user using an operation unit (not shown) of the mobile terminal to operate the power transmission request icon displayed on the display of the mobile terminal has

been received from the mobile terminal.

(78) When the power receiving side vehicle has received no power transmission request (NO in **S14**), the power receiving side vehicle returns the process to the main routine without performing contactless charging between the vehicles. On the other hand, when the power receiving side vehicle has received the power transmission request (YES in **S14**), the power receiving side vehicle outputs, to the power transmitting side vehicle, a request signal corresponding to the power transmission request and SOC information by which the SOC of battery **161** can be specified (**S15**).

(79) In response, upon receipt of the power transmission request and the SOC information from the power receiving side vehicle, the power transmitting side vehicle determines whether or not the power transmission permission condition is satisfied (**S21**). When the power transmission permission condition is not satisfied (NO in **S21**), the power transmitting side vehicle outputs, to the power receiving side vehicle, a prohibition signal corresponding to power transmission prohibition, and returns the process to the main routine.

(80) On the other hand, when the power transmission permission condition is satisfied (YES in **S21**), the power transmitting side vehicle performs a process of displaying the power transmission permission icon for receiving a power transmission permission from the user (**S23**). For example, ECU **11** of the power transmitting side vehicle displays the power transmission permission icon on display **132** of navigation system **13**. Alternatively, ECU **11** of the power transmitting side vehicle transmits an image signal for displaying the power transmission permission icon to the user's mobile terminal (not shown) such as a smartphone via communication module **14**. Thus, the power transmission permission icon is displayed on the display of the user's mobile terminal.

(81) The power transmitting side vehicle determines whether or not the power transmission permission by the user has been received (**S24**). For example, ECU **11** of the power transmitting side vehicle determines whether or not an instruction to permit power transmission by the user using operation unit **133** to operate the power transmission permission icon displayed on display **132** of navigation system **13** has been received from navigation system **13**. Alternatively, ECU **11** of the power transmitting side vehicle determines whether or not an instruction to permit power transmission by the user using an operation unit (not shown) of the mobile terminal to operate the power transmission permission icon displayed on the display of the mobile terminal has been received from the mobile terminal.

(82) It should be noted that the power transmitting side vehicle may be configured to allow the user to set in advance whether to permit power transmission in response to a power transmission request from a power receiving side vehicle. For example, the user of the power transmitting side vehicle may make a permission for a power transmission request in advance using operation unit **133**. In this case, ECU **11** of the power transmitting side vehicle may skip the processes of **S23** and **S24** and may perform a below-described process of **S25**.

(83) When the power transmission permission has not been received (NO in **S24**), the power transmitting side vehicle outputs a prohibition signal corresponding to power transmission prohibition to the power receiving side vehicle, and returns the process to the main routine. On the other hand, when the power transmission permission has been received (YES in **S24**), the power transmitting side vehicle outputs a permission signal corresponding to power transmission permission to the power receiving side vehicle (**S25**).

(84) In response, the power receiving side vehicle determines whether or not the permission signal has been received (**S16**). When no permission signal has been received from the power transmitting side vehicle, i.e., when the prohibition signal has been received from the power transmitting side vehicle (NO in **S16**), the power receiving side vehicle performs a power reception prohibiting process for prohibiting the power from being contactlessly received from the power transmitting side vehicle (**S17**), and returns the process to the main routine.

(85) On the other hand, when the permission signal has been received from the power transmitting

side vehicle (YES in S16), the power receiving side vehicle performs a power reception starting process for starting to contactlessly receive power from the power transmitting side vehicle (S18), and returns the process to the main routine. Thus, the power transmitting side vehicle contactlessly outputs, to the power receiving side vehicle, the power received by power receiving device 171, after charging battery 161 using the power received by power receiving device 171 from power transmitting facility 9 or without charging battery 161 using the power received by power receiving device 171 from power transmitting facility 9.

(86) It should be noted that the power transmitting side vehicle may stepwisely change the power to be output to the power receiving side vehicle, in accordance with the amount of power contactlessly received from power transmitting facility 9. That is, the power transmitting side vehicle may control power transmitting device 181 via converter 163 so as to contactlessly output, to the power receiving side vehicle, power corresponding to the amount of power contactlessly received from power transmitting facility 9.

(87) As described above, in the present embodiment, when the power receiving side vehicle cannot contactlessly receive power from power transmitting facility 9, the power receiving side vehicle can request the power transmitting side vehicle to contactlessly transmit power. When the power transmission request from the power receiving side vehicle is permitted, the power transmitting side vehicle can contactlessly output, to the power receiving side vehicle, the power from power transmitting facility 9 while contactlessly receiving the power from power transmitting facility 9. Thus, even when the power receiving side vehicle cannot contactlessly receive the power from power transmitting facility 9, battery 161 can be charged using the power from power transmitting facility 9 received via the power transmitting side vehicle. Therefore, the user's convenience is improved.

(88) Although the present disclosure has been described and illustrated in detail, it is clearly understood that the same is by way of illustration and example only and is not to be taken by way of limitation, the scope of the present disclosure being interpreted by the terms of the appended claims.

Claims

1. A charging system that contactlessly charges a vehicle with power, the charging system comprising: a first vehicle that stops to wait for a traffic light at a traveling lane leading to an intersection; a second vehicle that stops to wait for the traffic light at the traveling lane; and a power transmitting facility that contactlessly transmits power to each of the first vehicle and the second vehicle, the power transmitting facility being provided in the traveling lane, wherein when the first vehicle stops at a position at which the power from the power transmitting facility is unable to be contactlessly received and a distance between the first vehicle and the second vehicle is less than or equal to a distance in which power can be contactlessly transmitted from the second vehicle to the first vehicle, the first vehicle requests the second vehicle to output the power in accordance with an instruction input by the user of the first vehicle, when the second vehicle stops at a position at which the power from the power transmitting facility is able to be contactlessly received and a power transmission permission condition is satisfied, the second vehicle contactlessly outputs, to the first vehicle, the power received from the power transmitting facility in accordance with the request from the first vehicle, and the first vehicle contactlessly receives the power from the second vehicle.

2. The charging system according to claim 1, wherein when the first vehicle stops at the position at which the power from the power transmitting facility is unable to be contactlessly received and the distance between the first vehicle and the second vehicle is less than or equal to the distance in which power can be contactlessly transmitted from the second vehicle to the first vehicle, the first vehicle displays an icon for receiving the instruction from the user on a display of a navigation

system of the first vehicle or a mobile terminal of the user.

3. The charging system according to claim 2, wherein the power transmission permission condition includes a condition that the second vehicle is contactlessly receiving power from the power transmitting facility, a condition that an amount of power stored in the second vehicle is larger than an amount of power stored in the first vehicle, a condition that the amount of power stored in the second vehicle have reached an upper limit value or a condition that time of waiting for the traffic light at the intersection is more than or equal to a threshold value.

4. The charging system according to claim 2, wherein the second vehicle contactlessly outputs the power from the power transmitting facility to the first vehicle while charging a battery using the power from the power transmitting facility.

5. The charging system according to claim 2, wherein the second vehicle charges a battery using the power from the power transmitting facility, and contactlessly outputs the power stored in the battery to the first vehicle.

6. The charging system according to claim 2, wherein the second vehicle contactlessly outputs the power from the power transmitting facility to the first vehicle without charging a battery using the power from the power transmitting facility.
